

Is India Seriously Underestimated in China?

Zhihu

Judging by published articles, China's India scholars and the think-tank/strategic community have not underestimated India much. But the evaluation of India on Chinese self-media, Zhihu, WeChat, Douyin, and other platforms is definitely an underestimation.

For instance, a quick search on Zhihu turns up earnest "popular science" posts solemnly explaining that India has no fertiliser production capacity:

幻想成为联合国五常的印度，连化肥都造不出来



远方青木

公众号：远方青木 (ID: YFqingmu)

+ 关注

147 人赞同了该文章 >

联合国五常是世界上最强大的五个国家，而自认为是区域强国的印度经常提议让自己加入五常。

敢提议自己当联合国五常，印度实力即便不顶尖也不会太差吧。

但实际上，印度是一个连化肥都造不出来的国家。

印度是世界第一化肥进口大国，每年从全世界进口大量化肥，其中从中国进口的化肥数量就高达每年360万吨。

前几个月，印度还出现因化肥短缺导致农民抢购的新闻。

知乎 @最最遥远的路

But another quick search reveals that India not only has fertiliser production capacity — it is a major fertiliser producer, with annual output in the tens of millions of tonnes.

印度政府现在正计划对纳米磷酸二铵进行类似的推动，这样磷肥的消耗会减少，也不会影响作物产量。

印度当地所有肥料的产量增加了1.6%，达到3916.2万吨。其中，包括2320.2万吨的尿素、315万吨的磷酸二铵、819.4万吨的复合肥、405.7万吨的普钙和55.9万吨的硫酸铵。

据报道，在拉比季节的关键时期，印度各类化肥销售额均有增加，销量增长了21%。

Source: India's fertiliser imports fell 18% in FY2025

Of course, people will continue to object: India can't even make bullets, so how could it have the capacity to manufacture fertiliser production equipment? Its equipment is all bought!

But if you know how to search properly, you can actually find relevant reports. India does have the capacity to produce fertiliser plant equipment. India's leading company Larsen & Toubro, for instance, can manufacture all kinds of plant equipment, including fertiliser plant equipment. The original technology did not come from within the country, but India has mastered the manufacturing technology and is on the path of introduction–digestion–improvement.

Reference: Fertiliser | Process Plant | Products Services | L&T Heavy Engineering

Moreover, Larsen & Toubro is cooperating with the Japanese firm Itochu to develop an advanced green ammonia project — producing hydrogen by electrolysing water, then synthesising ammonia from hydrogen and nitrogen — a green, environmentally friendly technology for the fertiliser industry.

If the India in your eyes is at the level of the Nationalist government before liberation, then I'm sorry to say fertiliser is only the tip of the iceberg. You can write Zhihu answers mocking India however you like, but reality is reality; it will not actively conform to your wishes.

Let us together take a look at India's actual industrial level.

First, look at India's basic industry data:

Category	India (2024)	China (2000)	China (2004)	China (2010)
Steel	149.6 million t	128.5 million t	282.9 million t	595.6 million t
Electricity generation	1.824 trillion kWh	1.37 trillion kWh	2.187 trillion kWh	4.23 trillion kWh

Refining capacity	257 million t	210 million t	340 million t	617 million t
Cement output	450 million t	597 million t	970 million t	1.868 billion t
Fertiliser output	50.33 million t	29.554 million t	51 million t	66.198 million t
Passenger car output	4.27 million	2.0691 million	5.0705 million	13.8971 million
Expressway length	6,059 km (11,127 km under construction)	16,300 km	34,000 km	65,000 km
Coal output	1.047 billion t	1.3 billion t	1.956 billion t	3.24 billion t
Aluminium output	4.15 million t	2.8 million t	6.56 million t	15.65 million t

The other data you can look up for yourself; it is all much the same. These figures show that India's level of industrial development in 2024 roughly corresponds to China's level between 2000 and 2004.

Of course, some will still say India has no capacity to make its own equipment; it is all buy, buy, buy. First, the industrialisation of every late-developing country begins with buying — China included. Second, apart from buying, India is also developing core manufacturing capacity; it is just that Chinese media report little of it.

Take Larsen & Toubro again: besides fertiliser plant equipment, it also produces oil refining equipment, power generation equipment, and more.

Below is petrochemical equipment exported to Thailand:

Power plant equipment:

It even exports equipment to China:

Below is equipment exported to BASF's plant in Guangdong:

This next one was presumably imported by another Chinese company:

The Chinese often raise the objection: industrialisation needs electricity. China no longer exports coal power plants, so how will India generate electricity? Hence India cannot industrialise.

In fact, India does possess the capacity to manufacture coal power equipment.

India's BHEL commissioned a supercritical generating unit in 2021:

Source: <https://www.electricalindia.in/bhel-commissions-an-800-mw-supercritical-thermal-power-plant/>

China commissioned a generating unit of the same class in 2004. India trails China by nearly 20 years.

Then someone says: India can generate electricity, but it cannot manufacture transmission equipment; it has no UHV technology. In fact, India has this too.

Again the same company, BHEL, has manufactured 1200 kV AC UHV equipment.

Besides high-voltage AC transmission, this Indian company is cooperating with a Japanese firm on UHV DC transmission — 800 kV UHV DC, 1,200 kilometres long.

With electricity settled, someone will say India lacks steel; if China does not export steel-making equipment to it, India is finished.

Although Indian enterprises are relatively backward in this field, they can still make some equipment. For instance, Simplex Engineering & Foundry Works can manufacture blast furnace equipment. The technology may have been learned from China and other foreign countries originally, but being able to make relatively modern equipment shows a certain foundation exists.

Next, high-speed rail. In Chinese eyes, HSR is absolutely high tech; Indians certainly cannot master it. Not to mention HSR — even semi-high-speed, even ordinary trains, Indians cannot manage. What is the reality?

India produced its first quasi-high-speed train in 2019.

And it is developing a 250 km/h high-speed train.

China's first domestically developed 180 km/h train was produced in 1994; the first self-developed 250 km/h train was completed in 2002. Later China began importing advanced foreign technology and achieved a great leap in HSR. China is overall 20–30 years ahead of India, but India does have some foundation, and can likewise replicate China's HSR development path through introduction–absorption–innovation.

And last of all, India is even working on tunnel-boring machines — the TBMs used for metro and tunnel construction. China produced a 6-metre-diameter TBM in 2008. India currently plans first to make a 6.5-metre-diameter TBM; when it will be completed is unknown.

The well-known BEML company, which has a certain equipment manufacturing strength, is currently working on this project.

Project details can be found at: https://www.bemlindia.in/wp-content/plugins/tenders/WCP/DATA/uploads/CTPAM_EoI_TBM_Design%20Consultancy_2025.pdf

India's international environment is currently very good; and since Europe is being hit hard by Chinese TBMs, who knows — it may, before dying, sell India some mid- and low-end TBM

technology, accelerating India's development.

Finally, in the communications field, Indian enterprises have shown equal ambition.

Jio has developed its own 5G solution. Of course, it certainly contains a lot of foreign technology, but starting from assembly and gradually substituting domestic parts is itself the path of late-developing countries. Jio not only uses its own 5G technology domestically but plans to promote it in Africa.

The degree of Jio's genuine self-development is unclear. India also has a company called Tejas Networks, which can manufacture 4G RAN, LTE, and various other equipment. Its 5G equipment has presumably just begun and is probably not very good. However, since I myself am in the communications industry, the fact that India can make genuine 4G equipment is somewhat unexpected.

<https://www.tejasnetworks.com/technologies/4g-lte/>

<https://www.tejasnetworks.com/products/tejas-4g-radio-access-network/>

India is also developing its own 5G core network; the project is in the hands of a start-up, and the outcome will be known within a year or two.

As for India's chip industry chain, many people are following it, but since most of it is foreign-funded or joint-venture and there is not much indigenous technology, I will not discuss it here.

Through this introduction to India's basic industrial level and high-tech equipment manufacturing, one can glimpse India's overall level of development. India has great wealth disparity and uneven development, but its industrial base is being steadily consolidated and it is steadily climbing the ladder of advanced technology. The overall gap with China is some 20–25 years; loosely speaking, "20 years behind" is fair enough.

To view India as a hopeless country and completely deny India's industrial and technological achievements is not an objective approach — it is essentially no different from the way European and American media years ago imagined China as a shabby country with no high-rises.

I hope this article offers you a different perspective and inspires you to explore the real world on your own.

This article was originally published on Zhihu (in Chinese): <https://www.zhihu.com/question/349933016/answer/1979511765931942324> Author: 最遥远的路 · Zhihu profile: <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>

本文原发表于知乎专栏/回答（中文）： <https://www.zhihu.com/question/349933016/answer/1979511765931942324>；作者：最遥远的路： <https://www.zhihu.com/people/zui-zui-yao-yuan-de-lu-32/posts>